

Abhirup Saha

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🐙 GitHub profile

🌐 LinkedIn Profile

Education

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech in Computer Science and Technology; CGPA: 9.56/10 (upto 6th Semester)

Aug, 2023 – June, 2027

Howrah, WB, India

Kendriya Vidyalaya No.2, Saltlake

CBSE; AISSCE (12th) Score: 95%

April, 2023

Saltlake, WB, India

Kendriya Vidyalaya No.2, Saltlake

CBSE; AISSE (10th) Score: 96.4%

April, 2021

Saltlake, WB, India

Work Experience

RESEARCH INTERN at VECC — Nuclear Pulse Denoising 🔗

May, 2026 – July, 2026

- Architected a 4-stage physics-guided **Variational Autoencoder (VAE-PGNN)** pipeline to denoise nuclear pulse signals, achieving a **10× baseline noise reduction** and **+22 dB SNR improvement** on experimental radioisotope data.
- Synthesized a large-scale training dataset via **Monte Carlo domain randomization**, and benchmarked the VAE against classical filters (**Wiener, Kalman**) optimized using various **meta-heuristic** algorithms.

Projects

SMB Client App: Android app to access SMB file-shares 🔗

February, 2026 – Present

- Engineered a **Flutter (Dart)** Android application for accessing **SMBv2/3** network shares, implementing NTLM authentication and hierarchical file navigation.
- Secured user credentials with hardware-backed **AES-256 GCM** encryption via Android Keystore and mitigated MITM risks through strict local subnet IP validation.
- Optimized file operations by building an asynchronous multi-file download manager with in-memory **ZIP compression** and automated lifecycle-aware cache sanitization.

IPL Data Analysis and Winner Prediction 🔗

March, 2025

- Implemented **data cleaning** and **feature engineering** on IPL datasets, including the standardization of the dataset, extracting season identifiers, and deriving advanced cricket metrics.
- Conducted **EDA** using **statistical analysis** and clustering to identify key trends in team, player, and seasonal performance for **predictive ML model** development

Achievements

- Ranked **1st** among 29 teams in **Truth Tuner, 2025**, building an **ML model** for Indian News Bias Detection
- **Finalist** in **IESTxUCO Bank Hackathon** on **Fraud Detection in Banking Transactions**

Technical Skills

Languages: C, C++, Python, Dart, JavaScript, HTML, CSS, SQL

Frameworks and Libraries: Flutter, React, Selenium, NumPy, Pandas, Scikit-Learn, Matplotlib

Software Tools: Docker, Git, GitHub, Android Studio, VS Code, Linux

Core Subjects: OOPS, CN, OS, DBMS

Team Experience

LEAD . CORE MEMBER — Cybersecurity Chapter : CodeIEST, GDGoC IESTS

October 2025 – Present

- Successfully organized **RootAccess CTF, EOFool CTF** in AY 2025-26 with **300+** participants.
- Held multiple events on concepts of **Cryptography, Computer Networks**.